

The empirical recurrence for OEIS A297378, recovered from its tail

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Abstract

OEIS A297378 records “Empirical recurrence of order 84” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 212$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 88$. Its last coefficient is $c_{84} = 16 \neq 0$, so no further tail satisfies anything shorter; any order-84 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A297378 is “Number of $5 \times n$ 0..1 arrays with every 1 horizontally or antidiagonally adjacent to 0, 1 or 2 neighboring 1s.”. It has offset 1 and begins

32, 1024, 18805, 418344, 9554038, 211988289, 4729833008, 105725571407, ...

The entry states:

Empirical recurrence of order 84 (see link above)

As of the “Last modified” line on the live entry (Dec 29 2017 07:44:45, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 212$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 212$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 84: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 84.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 424$ exact terms from $n = 4$ on, it returns order 84 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{84} c_i a(n-i),$$

c_1	13	c_{22}	498565901194	c_{43}	−1836354742829	c_{64}	−5098215462
c_2	84	c_{23}	1748604848018	c_{44}	8969508976808	c_{65}	−267127888
c_3	2187	c_{24}	−2764350702842	c_{45}	13876730284408	c_{66}	−881152944
c_4	9502	c_{25}	−4641508691789	c_{46}	−1425244793440	c_{67}	108995567
c_5	79993	c_{26}	8482388798926	c_{47}	−1730336907866	c_{68}	−46792295
c_6	74082	c_{27}	10241009263996	c_{48}	−7035523945001	c_{69}	72196788
c_7	410747	c_{28}	−19067130931718	c_{49}	−224471113354	c_{70}	21166289
c_8	−5129649	c_{29}	−19640045590501	c_{50}	655901420094	c_{71}	14150847
c_9	−4433998	c_{30}	33624947064520	c_{51}	1038044946645	c_{72}	7620326
c_{10}	−45864443	c_{31}	33414962708142	c_{52}	842020277910	c_{73}	1904232
c_{11}	200415186	c_{32}	−47282731113692	c_{53}	−159683771588	c_{74}	894485
c_{12}	139182874	c_{33}	−50097522811668	c_{54}	440680558772	c_{75}	−61766
c_{13}	883440176	c_{34}	51991432252060	c_{55}	−122651279306	c_{76}	45883
c_{14}	−4575438044	c_{35}	64084717608886	c_{56}	119869216389	c_{77}	−46616
c_{15}	−6299434608	c_{36}	−41184185805547	c_{57}	−113326098040	c_{78}	−15800
c_{16}	29377432526	c_{37}	−66325222127829	c_{58}	−53269098167	c_{79}	−6552
c_{17}	3832159795	c_{38}	16384832152104	c_{59}	−61451847130	c_{80}	−1920
c_{18}	−82788097907	c_{39}	51373128672722	c_{60}	−43769947726	c_{81}	−192
c_{19}	81129921539	c_{40}	7674983233637	c_{61}	−18498507113	c_{82}	−192
c_{20}	67840939206	c_{41}	−20779172206414	c_{62}	−19066301762	c_{83}	32
c_{21}	−497885002881	c_{42}	−20332355231855	c_{63}	−4277657367	c_{84}	16

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 88$, and it is the recurrence OEIS A297378 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{84} - c_1x^{84-1} - \dots - c_{84}$. Its constant term is $-c_{84} = -16$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 84 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 84, so $\deg q = 84$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 84, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 84, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 16.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-84 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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